A Comparative Analysis of SVM Kernels using the Wine Dataset

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1. Introduction

Support Vector Machines (SVM) are widely used machine learning algorithms for classification tasks. They work by identifying a decision boundary that separates different classes in a dataset, using key data points known as support vectors. In many real-world situations, data is not linearly separable, meaning a straight line cannot effectively divide the classes. To address this, SVM uses kernel functions that transform the data into a higher-dimensional space, allowing the model to create more flexible decision boundaries. This makes SVM suitable for handling complex patterns. In this tutorial, three kernels—Linear, Polynomial, and Radial Basis Function (RBF)—are applied to the Wine dataset. The aim is to compare their performance and understand how each kernel affects classification results. By analysing both visualisations and accuracy, this study highlights the importance of selecting an appropriate kernel for effective machine learning models.

2. Dataset:

The dataset used in this study is the Wine dataset, obtained from the UCI Machine Learning Repository. It consists of 178 samples and 13 features representing chemical properties of wines derived from three different classes. These features include attributes such as alcohol content, magnesium, flavonoids, and colour intensity. The objective is to classify each sample into its correct class based on these characteristics. The dataset is well-structured and contains no missing values, making it suitable for machine learning tasks. Its moderate complexity and overlapping classes make it ideal for comparing the performance of different SVM kernels.

2.1 Data Preprocessing

Before applying the SVM models, the dataset was prepared using standard preprocessing techniques. First, feature scaling was performed using StandardScaler to ensure that all features contribute equally to the model, as SVM is sensitive to differences in feature magnitudes. Without scaling, features with larger values could dominate the model's performance. Next, Principal Component Analysis (PCA) was applied to reduce the dataset to two dimensions. This step helps in visualising the data and understanding how the classes are distributed. Although PCA reduces dimensionality, it retains the most important information required for effective analysis.

3. PCA Visualisation

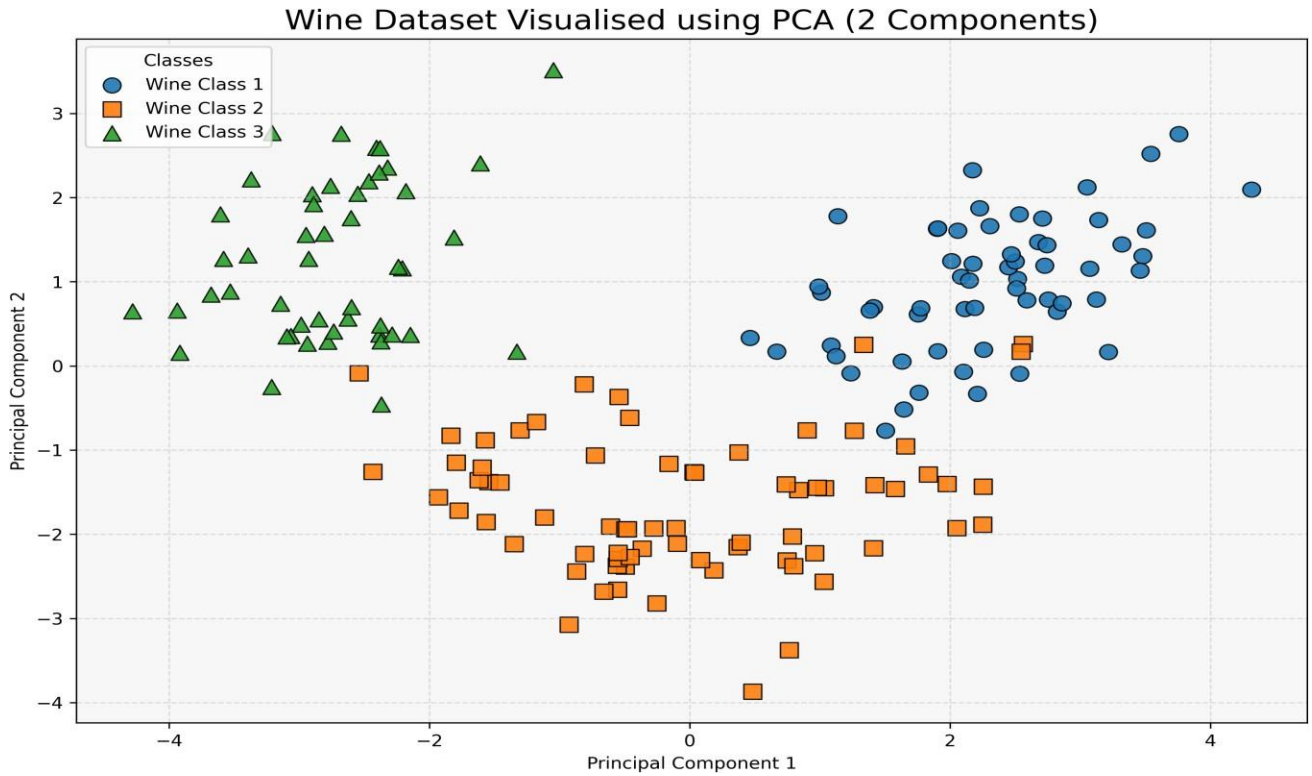

Fig 1: Wine dataset visualised using PCA

After applying PCA, the dataset was transformed into two principal components for visualisation. This allows the data to be plotted in a two-dimensional space, making it easier to observe the distribution of different classes. The PCA plot shows that the classes form distinct clusters, although some overlap is still present. This indicates that the dataset is not perfectly linearly separable, which can affect the performance of simple models. The visualisation provides useful insight into the structure of the data and highlights the need for more flexible models, such as SVM with non-linear kernels, to achieve better classification results.

4. SVM Models

4.1 Linear kernel Graph

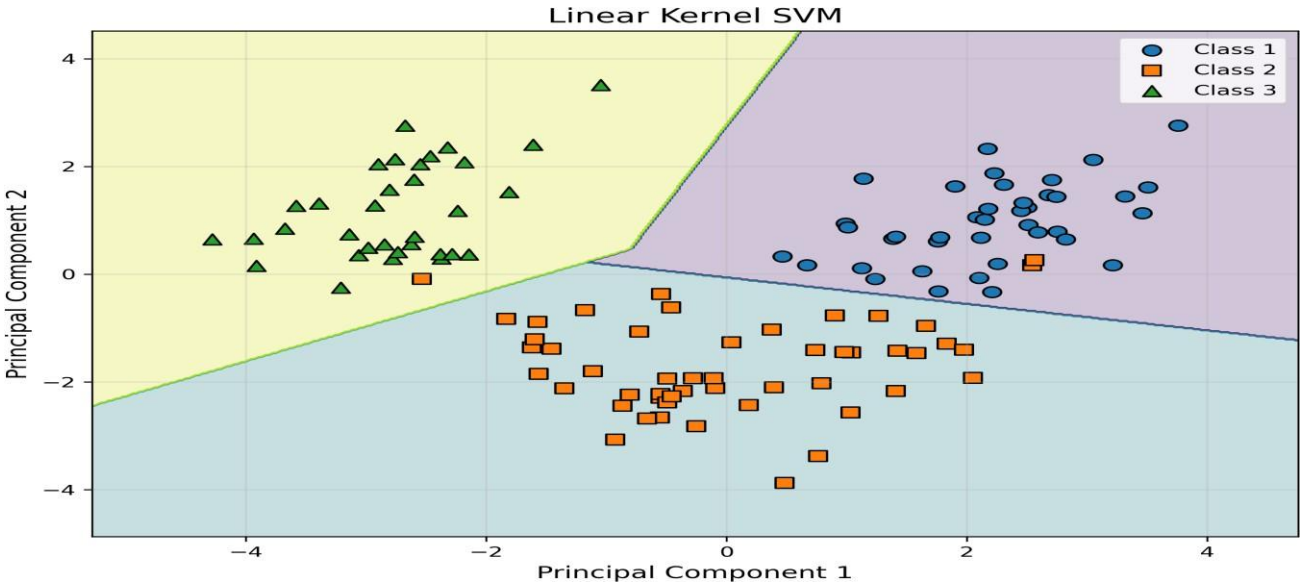

Fig 2: Decision Boundary using Linear Kernel

The linear kernel is the simplest form of SVM, where the model attempts to separate the classes using a straight decision boundary. In the graph, it can be observed that the boundary divides most of the data reasonably well. However, there are regions where the classes overlap, leading to some misclassification. This shows that the dataset is not perfectly linearly separable. While the linear kernel is computationally efficient and easy to interpret, it lacks the flexibility needed to handle more complex patterns in the data, which limits its overall performance compared to non-linear Kernels.

4.2 Polynomial Kernel Graph

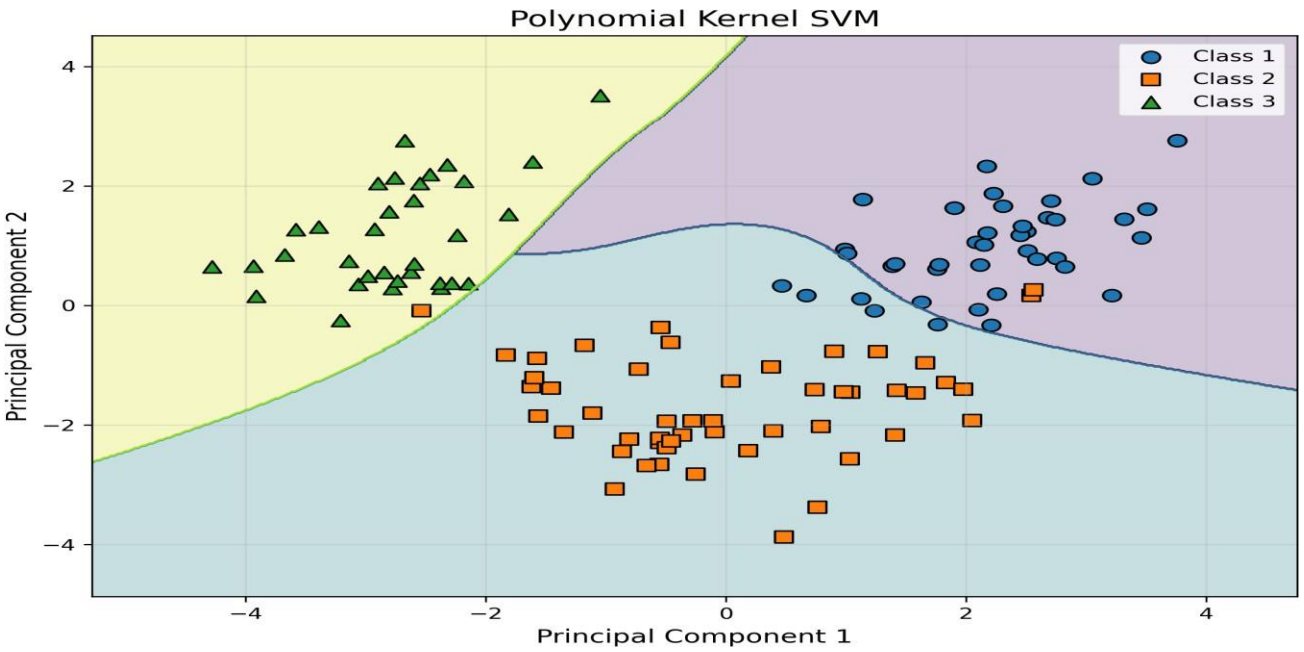

Fig 3. Decision boundary using Polynomial Kernel

The polynomial kernel introduces non-linearity by allowing the decision boundary to curve, enabling the model to capture more complex relationships between features. In the graph, the boundary appears more flexible compared to the linear kernel, which improves the separation of overlapping classes. This results in better classification performance. However, the polynomial kernel still has some limitations when dealing with highly complex data patterns. Its effectiveness also depends on the degree of the polynomial, which controls the level of flexibility. Overall, it provides an improvement over the linear kernel but may not always achieve the best results.

4.3 RBF Kernel Graph

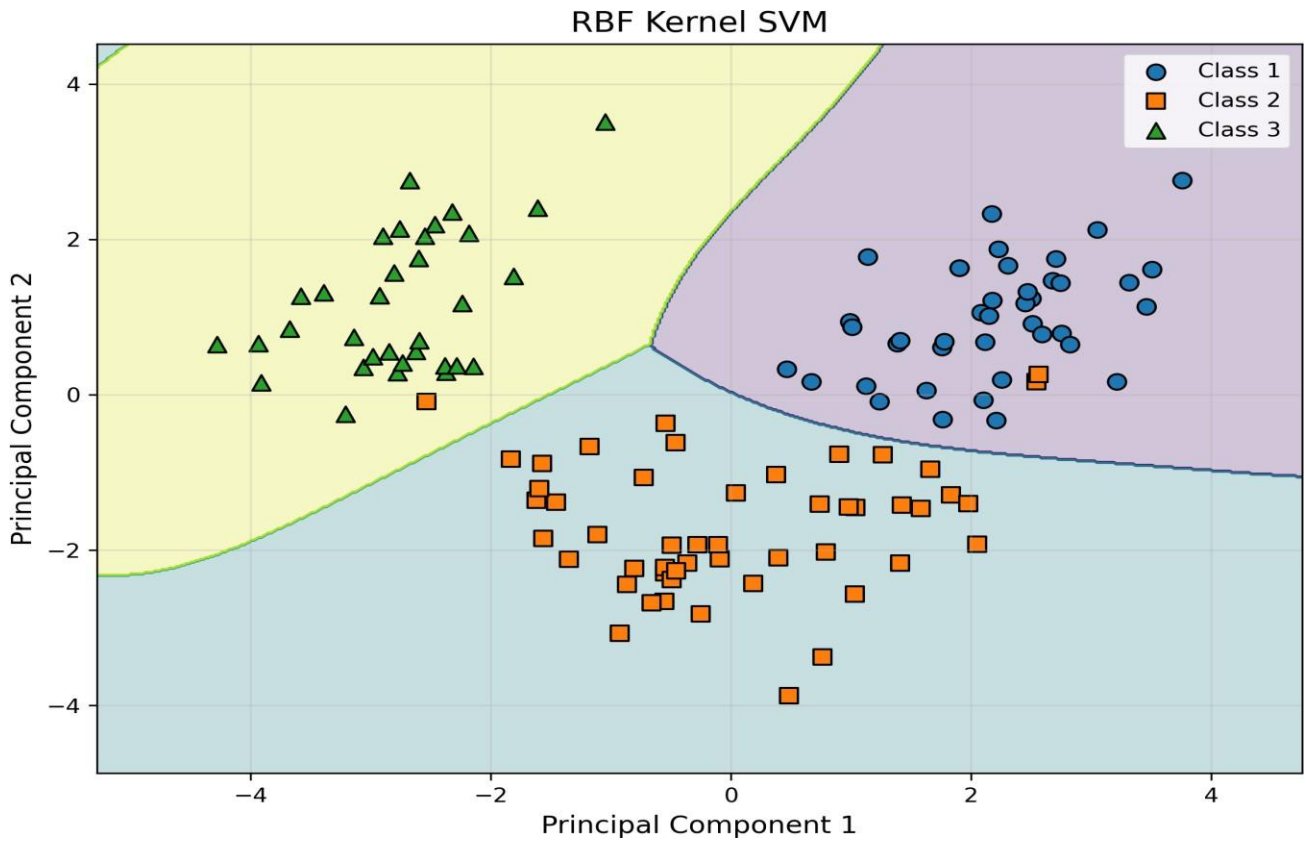

Fig 4: Decision Boundary using RBF Kernel

The Radial Basis Function (RBF) kernel is a highly flexible non-linear kernel that adapts well to the structure of the data. In the graph, the decision boundary closely follows the distribution of the data points, allowing for better separation of classes. This results in improved classification accuracy compared to both linear and polynomial kernels. The RBF kernel is particularly effective for datasets with complex patterns and overlapping regions. Its ability to map data into a higher-dimensional space makes it the most suitable choice in this case, as it provides the best overall performance among the tested models.

5. Model Evaluation

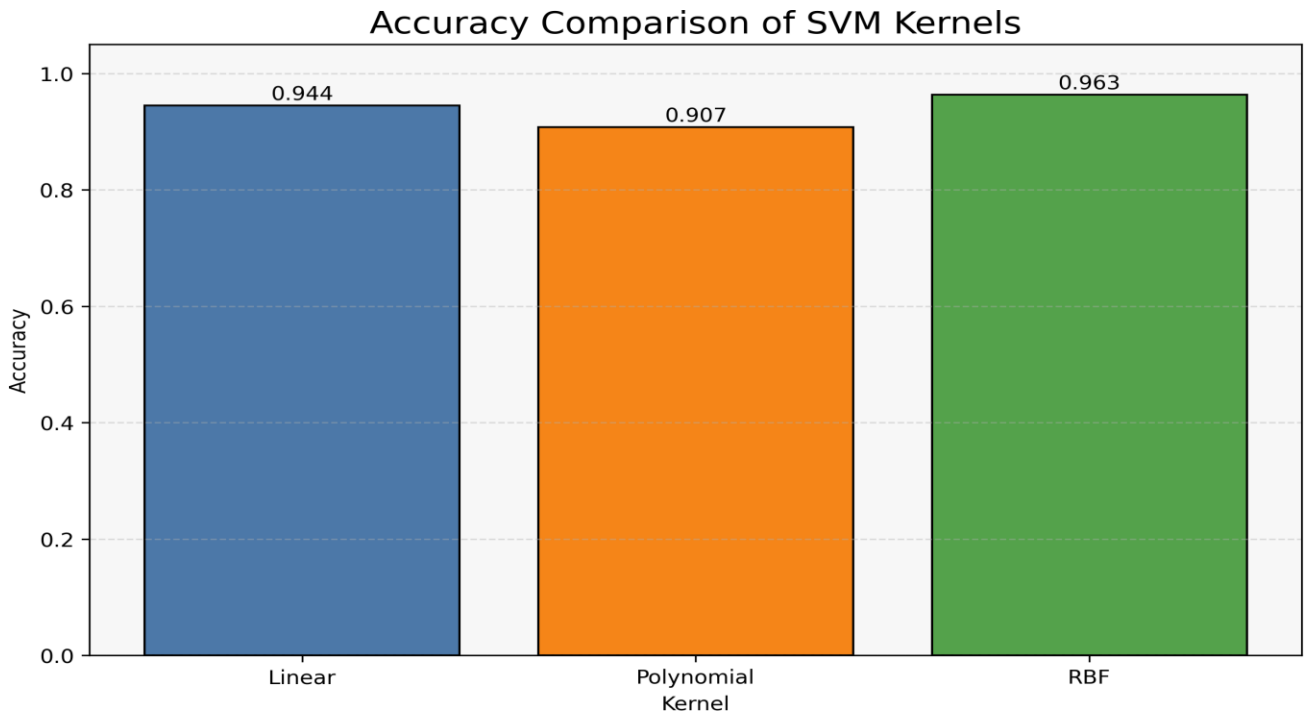

Fig 5: Accuracy comparison of SVM kernels

Kernel	Accuracy
Linear	0.94
Polynomial	0.90
RBF	0.96

The performance of the three SVM kernels was evaluated using classification accuracy. The results show that the linear kernel provides moderate accuracy, as it struggles to handle overlapping classes. The polynomial kernel improves performance by capturing some non-linear patterns, resulting in better separation of the data. However, the RBF kernel achieves the highest accuracy among all models. This is because it can adapt to complex data structures and create more flexible decision boundaries. The comparison clearly demonstrates that non-linear kernels, particularly RBF, are more effective for this dataset and provide better overall classification performance.

6. Discussion

The results clearly show that the choice of kernel has a significant impact on SVM performance. The linear kernel performs adequately but struggles in regions where the data is not linearly separable. The polynomial kernel improves the model by introducing non-linearity, allowing it to capture more complex relationships between features. However, it still faces

limitations in highly overlapping areas. The RBF kernel performs the best because of its ability to create highly flexible decision boundaries that closely follow the structure of the data. This makes it more suitable for handling complex datasets. Overall, the comparison highlights the importance of selecting an appropriate kernel to achieve better classification results. The performance differences between kernels highlight the importance of model selection in real-world machine learning applications

7. Ethical Considerations:

Machine learning models rely heavily on the quality and reliability of the data used for training. If the data contains bias, errors, or imbalances, the model may produce inaccurate or unfair predictions. Although the Wine dataset used in this study is well-structured and relatively balanced, real-world datasets can be more complex and may introduce ethical challenges. In critical applications such as healthcare, finance, or security, incorrect predictions could lead to serious consequences. Therefore, it is important to ensure that models are evaluated carefully, and that fairness, transparency, and reliability are considered when applying machine learning techniques.

8. Conclusion

This study explored the performance of different SVM kernels using the Wine dataset. The results showed that the linear kernel provides a simple solution but struggles with overlapping data. The polynomial kernel improves performance by capturing some non-linear patterns, while the RBF kernel achieves the best results due to its flexibility and ability to model complex relationships. The visualisations and accuracy comparison clearly highlight the impact of kernel selection on model performance. Overall, this study demonstrates that choosing an appropriate kernel is essential for achieving accurate and reliable classification results, especially when dealing with datasets that are not linearly separable.

9. References

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